CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 1: **A Square and a Cube** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of perfect squares between 200 and 900 (both excluded) is: 1

(a) 14 **(b)** 17 **(c)** 15 **(d)** 16

2. The smallest number to be added to 1234 to make it a perfect square is: 1

(a) 62 **(b)** 63 **(c)** 61 **(d)** 64

3. The largest number to be subtracted from 2577 to make it a perfect square is: 1

(a) 79 **(b)** 77 **(c)** 80 **(d)** 78

4. The difference of the squares of two consecutive natural numbers is 59. The numbers are: 1

(a) 28 and 29 **(b)** 29 and 30 **(c)** 30 and 31 **(d)** 29 and 31

5. The number of digits in the square root of a 5-digit perfect square is: 1

(a) 2 **(b)** 3 **(c)** 4 **(d)** 5

6. $\sqrt[3]{-729}$ is: 1

(a) -9 **(b)** 10 **(c)** -10 **(d)** 9

7. The ones digit of 127^3 is: 1

(a) 9 **(b)** 7 **(c)** 1 **(d)** 3

8. The value of $\sqrt{196} + \sqrt{361}$ is: 1

(a) 34 **(b)** 266 **(c)** 5 **(d)** 33

9. **Assertion (A):** The number 1296 is a perfect square. 1

Reason (R): A number is a perfect square if every prime in its prime factorisation occurs an even number of times.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The square of an even number is always even. 1

Reason (R): The sum of the first n odd natural numbers is n^2.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the smallest number to be added to 3900 to make it a perfect square. What is the square root of the result? **2**

12. The difference between the squares of two consecutive numbers is 75. Find the numbers. **2**

13. Find the number of non-square numbers between 6^2 and 9^2. **2**

14. Find the smallest number to be added to 5680 to make it a perfect square. What is the square root of the result? **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A square garden has area 1225 m². Fencing costs ₹40 per metre. Find the total cost of fencing it. **3**

16. A solid cube of side 6 cm is cut into small cubes of side 1 cm. How many small cubes are formed? How many have exactly three painted faces if the big cube was painted all over? **3**

17. A rectangular hall 36 m by 48 m is to be paved with equal square tiles, the largest possible. Find the side of each tile and the number of tiles. **3**

OR

A square garden has area 2304 m². Fencing costs ₹35 per metre. Find the total cost of fencing it. **3**

18. A solid cube of side 7 cm is cut into small cubes of side 1 cm. How many small cubes are formed? How many have exactly three painted faces if the big cube was painted all over? **3**

OR

A rectangular hall 42 m by 56 m is to be paved with equal square tiles, the largest possible. Find the side of each tile and the number of tiles. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Ramesh, a gardener in Jaipur, is given a square plot of area 1764 m^2 to plant a lawn and fence it. He wants to find its dimensions using prime factorisation. **5**

(i) Express 1764 as a product of prime factors.

(ii) Find the side of the plot.

(iii) Find the cost of fencing all four sides at ₹50 per metre.

(iv) The owner decides to increase the side to 45 m. Find the extra area added.

20. Find the smallest square number that is divisible by each of 4, 9 and 15. **5**

OR

Find the smallest cube number that is divisible by each of 6, 10 and 15. **5**

END OF QUESTION PAPER — Total: 40 Marks